

PRAFFUL GUPTA

Backend Engineer | Scalable REST APIs | Distributed Systems | Database Optimization

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TECHNICAL SKILLS

- **Languages:** Java, Python, SQL
- **Backend Frameworks:** Spring Boot, Dropwizard, FastAPI, REST APIs
- **Databases & Caching:** MySQL, PostgreSQL, Redis, DuckDB, Spring Data JPA
- **Tools:** Git, GitHub, Docker, IntelliJ IDEA, VS Code, Postman
- **Core Concepts:** Distributed Systems, Rate Limiting, Database Indexing, Query Optimization, Transaction Management, Data Structures & Algorithms

WORK EXPERIENCE

Backend Developer Intern

May 2025 - Nov 2025

Mind Bridge Developers

- Engineered scalable REST APIs using **Java** and **Spring Boot** for an **ERP platform** serving **1,000+ users**.
- Eliminated duplicate payments under concurrent requests by enforcing **transactional integrity** and optimizing **database transaction boundaries**.
- Resolved **N+1 query bottlenecks** and implemented targeted database indexing, reducing report latency by **~60%**.
- Reduced high-volume API latency from 2-3s to **<800ms** through SQL query optimization, indexing improvements, and backend refactoring.

RESEARCH & PATENTS

Predicting Code Smell Severity in Large Open-Source Systems

First author, with S. Singh and P. K. Singh (KIET) — under review, IEEE conference

- Built a reproducible severity-prediction pipeline over **6,907** consensus-labelled code smell instances from **326** Apache and Eclipse repositories, benchmarking **11** ML algorithms under severe class imbalance.
- Raised best four-class macro-F1 from **0.449 to 0.489** and binary ROC-AUC from **0.884 to 0.942** by deriving complexity features directly from source at each instance's pinned commit.
- Validated under a strict cross-project protocol (no repository shared between train and test): mean macro-F1 drop of only **0.016**, showing the model learns general properties of complex code rather than project-specific idiosyncrasy.
- Reported all null results with Benjamini-Hochberg correction; SHAP analysis attributes predictions chiefly to code size and class shape. [GitHub](#)

AI-Based Pest Detection and Automated Pesticide Administration Device

Indian Patent Office, Application No. 202611057250 — filed 05 May 2026, published 17 July 2026

- Computer vision and machine learning system identifying pest species from insect images and leaf-damage patterns, integrated with IoT hardware for automated, dose-controlled spraying determined by pest type and crop growth stage.

Antidepressant Structure-Toxicity Signal Validation Using ICU Electronic Health Records

In progress — PhysioNet credentialed access to MIMIC-IV (MIT Laboratory for Computational Physiology)

- Testing whether drug-safety signals derived from FDA FAERS spontaneous reports reproduce in measured clinical data across **546,000+** hospital admissions; built a tested, reproducible DuckDB extraction pipeline with runtime lab-code resolution and dose-based exposure stratification.

PROJECTS

AutoDeploy-AI — CI/CD Build-Failure Prediction

Python, scikit-learn, SHAP, FastAPI, GitHub API

[GitHub](#)

- Built an end-to-end CI build-failure prediction system over **261k builds** from **243** open-source projects (TravisTorrent), engineering **31** change, history, and process features under strict **temporal leakage prevention** — every history feature computed causally from data available before the build, verified by manual recomputation.
- Evaluated on a **held-out-projects split** alongside a temporal split rather than a random split alone: **PR-AUC 0.80** temporal and **0.69** on previously unseen repositories, **~0.24** above a previous-build-outcome baseline. SHAP attribution concentrated ~66% of feature importance in three build-history autocorrelation features.
- Shipped a **FastAPI** prediction service with a live GitHub feature extractor, then identified and documented **train-serve skew** before release — trained on Travis CI outcomes, serving against GitHub Actions — producing a feature-parity report classifying which of the 31 features transfer exactly, approximately, or not at all, and adding an explicit cold-start state that reports low confidence on new repositories rather than over-flagging them.

Real-Time Collaborative Document Editor

Spring Boot, WebSockets, PostgreSQL, Vercel, Render

[GitHub](#) | [Live Demo](#)

- Developed a **WebSocket-based** collaborative editor supporting multi-user editing with **<100 ms sync latency**.
- Designed a **microservice architecture** (API Gateway + Persistence Service) for scalable real-time collaboration.
- Implemented operation-based edit propagation with version tracking to maintain consistent document state across concurrent clients.

Distributed Rate Limiting System

FastAPI, Redis, Lua, Docker, Render

[GitHub](#) | [Live Demo](#)

- Built a distributed rate limiter using **Redis (token bucket)** handling **10k+ req/min** with atomic throttling.
- Ensured consistent rate enforcement across distributed API instances using Redis atomic operations under high concurrency.
- Implemented configurable per-user and per-endpoint rate policies to prevent burst traffic and API abuse.

EDUCATION

Master of Computer Applications (MCA), pursuing | CGPA: 8.22 / 10

KIET Deemed to be University, Delhi-NCR · 2025 - 2027

Bachelor of Science (Life Sciences) | CGPA: 7.04 / 10

CCS University, Meerut · 2021 - 2024

CERTIFICATIONS

- **CITI Program — Data or Specimens Only Research** (Human Subjects Research), 2026. Credentialed by PhysioNet (MIT Laboratory for Computational Physiology) for restricted-access clinical databases.